



East West University
Dept. of Mathematical and Physical
Sciences (MPS)
Semester: SPRING 2025

Course Outline of PHY109

Course Code : PHY 109
Pre-requisite : None
Credit Hours : 3(Theory) +1(Lab) = 4
Course Title : Engineering Physics- I (Introductory Classical Physics)
Section : **10**

Course Instructor : Aswariya Roy
E-mail : aswariya.roy@ewubd.edu

Class Time :

Theory:

Day	Time	Room
Monday	1:30PM - 3:00PM	Room : FUB-601
Wednesday	1:30PM - 3:00PM	Room : FUB-504

Classroom Code : kcqyzuv

Objective of the course:

Physics is the subject that deals about the principles of laws of matter and energy. To build up a career in engineering discipline it is very important to have knowledge of Physics. Physics is the basic/fundamental subject in science and engineering. Physics plays always important role in developing of new technology. The course Engineering Physics-I has been designed to give a clear concept in the area of mechanics, fluid dynamics, SHM, waves (mechanical and electromagnetic), thermal processes and optics. All these areas are basic/fundamental in Physics. The concept in these areas can help the students for studying different engineering and communication courses.

Course Learning Outcomes:

After completing this PHY109 course students will be able to

1. apply the concept of force in their higher courses.
2. understand the fluid flow and concepts of Continuity and Bernoulli's equations
3. know the different types of waves and how wave convey energy.
4. know thermodynamic laws and entropy and ideal heat engine.
5. know about light and interference of light, diffraction of light and polarization of light.
6. know about the concept of electric and magnetic field and different laws related to electro-magnetism.

Course Contents /Descriptions:

1. Mechanics: Newton's law of motion; Kinetic energy, potential energy and work; Momentum, impulse, Collisions; Rotation of rigid body; Rotational dynamics; Equilibrium of rigid bodies.

2. Fluid Mechanics: Concept of fluids; Pressure and density; Measurement of pressure; General concept of fluid flow; The equation of continuity; Bernoulli's equation; Applications.

3. Oscillations and Waves: Simple harmonic oscillation, energy in SHM, damped harmonic motion; Different types of waves; The superposition principle; Wave speed; Energy and Power of traveling wave along string; Interference of waves; standing waves and resonance.

4. Thermal Physics: Concept of temperature and heat; isothermal and adiabatic changes; Reversible and irreversible processes; Three laws of thermodynamics; Thermodynamics and the concept of entropy, Heat engine.

5. Electricity and Magnetism: Concept of electric charge, the electric field, dipole in an electric field, electric flux, Gauss' law, electric potential, calculation of capacitance, capacitors with dielectric, energy storage in an electric field. The magnetic field, Biot-Savart law, Ampere's law, magnetic force on a current, magnetic lines of induction, Faraday's law, Lenz's Law.

6. Physical Optics: Light as electromagnetic wave; interference, diffraction of light, polarization of light; Young's double slit experiment; Michelson interferometer, Poynting vector.

Text Book:

1. Fundamentals of Physics; Halliday, Resnick and Walker, 10th Edition, Wiley & Sons.

Reference book:

1. University Physics; Sears and Zemansky's, Hugh D. Young, Roger A. Freedman, Pearson Education, Inc. London, UK.
2. Mechanics, Wave Motion & Heat; Sears, F.W.; Addison Wesley Publishing Company
3. Electricity and Magnetism, Dakworth, H. E.
4. Introduction to Optics, By Frank L. Pedrotti, Leno M. Pedrotti, Leno S. Pedrotti

Assessment/Evaluation:

Test/Exam	% of Marks
Class Tests/Quiz	10
Class Attendance/ Performance	10
Assignment/Presentation	5
Lab Work	25
Mid Term Exam	20
Final Exam	30

Grading Policy:

Marks (%)	Letter Grade	Grade Point
80- 100	A+	4.0
75-below 80	A	3.75
70-below 75	A-	3.5
65-below 70	B+	3.25
60-below 65	B	3.0
55-below 60	B-	2.75
50-below 55	C+	2.50
45-below 50	C	2.25
40-below 45	D	2.00
Below 40	F	0

Special Instructions:

- There will be no make-up quizzes
- Make-up exams for Term1 and Term 2 may be taken in case of valid reason. Student has to apply for make-up exam with proper document.
- Regular class attendance will be taken. The students are advised to attend classes regularly and keep regular contact with instructor for academic problems.
- There is zero tolerance for cheating at EWU. Students caught with cheat sheet in their possession, whether used or not used, &/ or copying from cheat sheets, writings on the palm of hand, back of calculator, chairs or nearby walls, etc. would be treated as cheating in the exam hall. The only penalty for cheating is expulsion.

Date: 16th February, 2025